

Companion XIV

Primordial Black Holes in the Relaxation-Driven Cyclic Cosmology

Relaxation-Suppressed Collapse, Sectoral Asymmetry, and Early-Universe Constraints

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April 18, 2026

Abstract

Primordial black holes (PBHs) provide a unique probe of the early Universe, sensitive to the thermal history, the growth of perturbations, and the collapse thresholds during radiation domination. Within the Relaxation-Driven Cyclic Cosmology (RDCC), the bipartite CPT-symmetric global state and the universal relaxation mechanism fundamentally alter the conditions for PBH formation. In this Companion we show that the relaxation dynamics introduced in Companion X naturally suppress the formation of PBHs in the visible sector by increasing the effective collapse threshold and modifying the growth of super-horizon perturbations. The temperature asymmetry between the two sectors leads to a sectoral PBH asymmetry, with the CPT-conjugate sector generically exhibiting a higher PBH abundance. We derive the modified collapse criterion, compute the relaxation-corrected mass function, and identify the conditions under which PBHs are strongly suppressed, mildly present, or entirely absent in the visible sector. The framework yields concrete predictions for the stochastic gravitational-wave background, microlensing constraints, and the PBH mass spectrum. These results close the third major open gap in the RDCC program and establish PBHs as a sensitive probe of relaxation-driven early-Universe physics.

1 Motivation

Primordial black holes (PBHs) occupy a unique position in modern cosmology. Unlike astrophysical black holes, which form from the collapse of massive stars, PBHs originate from the earliest moments of the Universe and therefore provide a direct probe of the physics operating near the horizon scale during radiation domination. Their abundance, mass spectrum, and formation conditions are sensitive to the thermal history, the growth of primordial perturbations, and the collapse thresholds that govern the transition from overdensity to black hole formation.

Within the standard cosmological framework, PBHs require either enhanced primordial fluctuations, non-Gaussian features, or exotic early-Universe dynamics to form in appreciable numbers. As a result, PBHs have long been regarded as a potential window into physics beyond the standard model of cosmology. Observational constraints from microlensing, the cosmic microwave background, and the stochastic gravitational-wave background have significantly restricted the viable parameter space, but the possibility of PBHs as dark matter or as sources of gravitational-wave events remains an active area of research.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The bipartite CPT-symmetric global state and the universal relaxation mechanism introduced in Companion X modify the early-Universe dynamics in ways that directly impact PBH formation. The relaxation rate alters the growth of super-horizon perturbations, increases the effective collapse threshold, and introduces a sectoral asymmetry due to the temperature difference between the two sectors. These effects naturally suppress PBH formation in the visible sector without requiring fine-tuning or additional fields.

The purpose of this Companion is to develop the PBH sector within RDCC, closing one of the remaining major gaps in the framework. We derive the relaxation-corrected collapse criterion, compute the modified PBH mass function, and identify the conditions under which PBHs are suppressed, mildly present, or entirely absent in the visible sector. We also show that the CPT-conjugate sector generically exhibits a higher PBH abundance, providing a distinctive signature of the bipartite global state.

Understanding PBH formation in RDCC is essential for three reasons:

- **Early-Universe consistency:** PBHs are sensitive to the thermal and dynamical history of the Universe, making them a natural probe of relaxation-driven cosmology.
- **Sectoral structure:** The temperature asymmetry between the two sectors implies different collapse thresholds and mass functions, offering a direct test of the bipartite global state.
- **Observational leverage:** PBHs leave imprints in gravitational waves, microlensing surveys, and the matter power spectrum, providing multiple avenues for testing RDCC predictions.

This Companion establishes PBHs as a sensitive and testable component of the RDCC framework and provides the theoretical foundation for interpreting PBH-related observables within relaxation-driven cosmology.

2 Primordial Black Holes in Standard Cosmology

Primordial black holes (PBHs) arise in standard cosmology when large density fluctuations re-enter the horizon during radiation domination and collapse under their own gravity. The formation criterion is typically expressed in terms of the density contrast δ exceeding a critical threshold δ_c , which depends on the equation of state and the shape of the perturbation. For a radiation-dominated Universe, numerical studies yield a collapse threshold of order

$$\delta_c \simeq 0.4 - 0.5, \tag{1}$$

with the precise value depending on the perturbation profile.

In the standard framework, PBH formation requires either:

- enhanced primordial fluctuations (e.g. from inflationary features),
- non-Gaussian statistics,
- or exotic early-Universe dynamics that temporarily reduce the collapse threshold.

Absent such enhancements, PBHs are exponentially suppressed due to the Gaussian tail of the primordial fluctuation distribution. The resulting PBH abundance is therefore highly sensitive to the amplitude and shape of the primordial power spectrum.

The PBH mass is set by the horizon mass at the time of collapse,

$$M_{\text{PBH}} \sim M_{\text{H}}(a_{\text{coll}}), \tag{2}$$

leading to a broad range of possible masses, from sub-asteroidal scales to many solar masses. Observational constraints from microlensing, the cosmic microwave background, and the stochastic gravitational-wave background significantly restrict the allowed PBH abundance, but do not exclude the possibility of PBHs contributing to dark matter or to gravitational-wave events observed by LIGO/Virgo/KAGRA.

While this standard picture provides a well-defined framework for PBH formation, it relies critically on the thermal history and perturbation growth during radiation domination. Any modification to the early-Universe dynamics—such as changes in the effective equation of state, entropy flow, or super-horizon evolution—can dramatically alter the PBH abundance.

In the Relaxation-Driven Cyclic Cosmology (RDCC), precisely such modifications arise naturally. The relaxation mechanism introduced in Companion X alters the growth of super-horizon perturbations, increases the effective collapse threshold, and introduces a sectoral asymmetry due to the temperature difference between the two sectors. These effects fundamentally change the conditions for PBH formation, as we develop in the following sections.

3 RDCC Framework: Early-Universe Relaxation and PBH Formation

The Relaxation-Driven Cyclic Cosmology (RDCC) is built upon a bipartite, CPT-symmetric global state in which two thermodynamically distinct sectors evolve on the same spacetime geometry. The observable Universe corresponds to a projection onto the sector with a forward thermodynamic arrow, while the CPT-conjugate sector evolves with decreasing entropy. This structure ensures global CPT invariance while allowing for emergent time asymmetry within each sector.

A central ingredient of RDCC is the universal relaxation mechanism introduced in Companion X. The relaxation rate $\Gamma(a)$ governs the flow of information and entropy between the two sectors and determines when sector-specific quantities freeze out. During the early Universe, when the scale factor is small, the relaxation rate satisfies

$$\Gamma(a) \gg H(a), \quad (3)$$

implying that the two sectors remain strongly entangled and behave effectively as a single CPT-symmetric system. As the Universe expands, the relaxation rate decreases according to the scaling law derived in Companion X,

$$\Gamma(a) = \alpha(a) H(a), \quad (4)$$

where $\alpha(a)$ is a slowly varying dimensionless relaxation coefficient. The transition $\Gamma(a) \simeq H(a)$ marks the onset of sectoral decoherence and the emergence of sector-specific effective dynamics.

This relaxation dynamics has direct implications for primordial black hole (PBH) formation. PBHs form when large density fluctuations re-enter the horizon during radiation domination and collapse under their own gravity. The collapse threshold and the growth of super-horizon perturbations are highly sensitive to the early-Universe dynamics. In RDCC, two effects are particularly important:

- **Relaxation-modified perturbation growth:** Super-horizon perturbations evolve differently when the relaxation rate is comparable to the Hubble rate. The effective equation of motion acquires a relaxation-induced damping term, suppressing the growth of overdensities prior to horizon re-entry.
- **Sectoral temperature asymmetry:** The two sectors possess different temperatures, with the ratio $T_B/T_A \simeq 0.5$ derived in Companion X. This modifies the sound speed, the horizon mass, and the collapse threshold in each sector, leading to a sectoral asymmetry in PBH formation.

Together, these effects increase the effective collapse threshold in the visible sector and suppress PBH formation relative to standard cosmology. The CPT-conjugate sector, being colder and possessing a different relaxation history, generically exhibits a higher PBH abundance. This sectoral asymmetry is a distinctive prediction of RDCC and provides a powerful observational handle on the bipartite global state.

In the following sections, we derive the relaxation-corrected collapse criterion, compute the modified PBH mass function, and analyze the sectoral PBH asymmetry implied by the RDCC framework.

4 Modified Collapse Threshold in RDCC

Primordial black holes form when large density fluctuations re-enter the horizon during radiation domination and exceed a critical collapse threshold δ_c . In standard cosmology, this threshold is determined by the competition between gravitational attraction and pressure support, yielding values of order $\delta_c \simeq 0.4 - 0.5$. In the Relaxation-Driven Cyclic Cosmology (RDCC), the early-Universe dynamics are modified by the universal relaxation mechanism introduced in Companion X, leading to a relaxation-corrected collapse criterion.

The key observation is that the relaxation rate $\Gamma(a)$ introduces an additional damping term in the evolution of super-horizon perturbations. For a density contrast $\delta(a)$, the effective equation of motion becomes

$$\ddot{\delta} + (2H + \Gamma) \dot{\delta} - c_s^2 \frac{k^2}{a^2} \delta = 0, \quad (5)$$

where c_s is the sound speed. When $\Gamma \gg H$, the damping term suppresses the growth of perturbations, preventing them from reaching the standard collapse threshold by the time they re-enter the horizon. As the Universe expands and Γ decreases, the suppression weakens, but the early-time damping leaves a lasting imprint on the amplitude of δ .

To quantify this effect, we define the relaxation-corrected collapse threshold δ_c^{RDCC} as the minimum amplitude at horizon re-entry required for collapse in the presence of relaxation-induced damping. Solving the perturbation equation in the regime $\Gamma \sim H$ yields an effective enhancement of the threshold,

$$\delta_c^{\text{RDCC}} = \delta_c^{\text{std}} \left[1 + \beta \frac{\Gamma}{H} \right], \quad (6)$$

where β is a dimensionless coefficient of order unity determined by the perturbation profile. Since $\Gamma/H = \alpha(a)$, with $\alpha(a)$ slowly varying and of order $\mathcal{O}(0.1 - 1)$ during the relevant epoch, the collapse threshold is generically increased by tens of percent.

This enhancement has two important consequences:

- **Exponential suppression of PBH formation:** PBH formation is exponentially sensitive to the collapse threshold. Even a modest increase in δ_c leads to a dramatic reduction in the PBH abundance in the visible sector.
- **Sectoral asymmetry:** The CPT-conjugate sector has a lower temperature and a different relaxation history, implying a smaller effective Γ/H ratio. As a result, the collapse threshold in the mirror sector is lower,

$$\delta_c^{(B)} < \delta_c^{(A)}, \quad (7)$$

leading to a higher PBH abundance in sector B.

The relaxation-corrected collapse threshold is therefore a natural and robust mechanism for suppressing PBH formation in the visible sector. No fine-tuning or additional fields are required; the suppression arises directly from the universal relaxation dynamics of RDCC.

In the next section, we analyze the resulting sectoral PBH asymmetry in more detail and derive the implications for the PBH mass spectrum in both sectors.

5 Sectoral PBH Asymmetry

A distinctive feature of the Relaxation-Driven Cyclic Cosmology (RDCC) is the bipartite CPT-symmetric global state, in which the two sectors evolve with opposite thermodynamic arrows and possess different effective temperatures. As shown in Companion X, the temperature ratio between the sectors satisfies

$$\frac{T_B}{T_A} \simeq 0.5, \quad (8)$$

implying that the CPT-conjugate sector is colder throughout the early Universe. This temperature asymmetry, combined with the relaxation-modified perturbation dynamics, leads to a natural and robust sectoral asymmetry in primordial black hole (PBH) formation.

PBH formation is highly sensitive to both the collapse threshold and the horizon mass at the time of collapse. Since the collapse threshold in RDCC is enhanced by relaxation-induced damping,

$$\delta_c^{(A)} = \delta_c^{\text{std}} [1 + \beta \alpha_A(a)], \quad (9)$$

the visible sector (sector A) experiences a stronger suppression of PBH formation. In the CPT-conjugate sector, the relaxation coefficient is smaller due to the lower temperature,

$$\alpha_B(a) < \alpha_A(a), \quad (10)$$

leading to a reduced collapse threshold,

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (11)$$

This difference is amplified by the exponential sensitivity of PBH formation to the collapse threshold. For Gaussian primordial fluctuations with variance σ^2 , the PBH abundance scales as

$$\beta_{\text{PBH}} \propto \exp \left[-\frac{\delta_c^2}{2\sigma^2} \right]. \quad (12)$$

A modest reduction in δ_c in sector B therefore leads to an exponentially larger PBH abundance relative to sector A.

In addition to the collapse threshold, the horizon mass at the time of collapse differs between the sectors due to the temperature asymmetry. The horizon mass scales as

$$M_H \propto g_*^{-1/2} T^{-2}, \quad (13)$$

implying that the colder sector B has a larger horizon mass at a given scale factor. As a result, PBHs formed in sector B are generically heavier than those formed in sector A.

Combining these effects yields three robust predictions:

- **Suppressed PBH abundance in the visible sector:** The enhanced collapse threshold and reduced horizon mass in sector A strongly suppress PBH formation relative to standard cosmology.
- **Enhanced PBH abundance in the CPT-conjugate sector:** The colder temperature and reduced relaxation-induced damping in sector B lead to a lower collapse threshold and a higher PBH abundance.
- **Sectoral mass asymmetry:** PBHs in sector B are generically heavier due to the larger horizon mass at collapse, providing a distinctive signature of the bipartite global state.

This sectoral PBH asymmetry is a direct consequence of the RDCC framework and requires no additional assumptions or parameters. It provides a powerful observational handle on the bipartite structure of the global state and the relaxation dynamics that govern the early Universe.

In the next section, we compute the relaxation-corrected PBH mass function and quantify the degree of suppression in the visible sector.

6 Relaxation-Corrected PBH Mass Function

The primordial black hole (PBH) mass function is determined by the abundance of density fluctuations that exceed the collapse threshold at horizon re-entry. In standard cosmology, the PBH mass fraction at formation is given by

$$\beta_{\text{PBH}}(M) = \int_{\delta_c}^{\infty} \frac{1}{\sqrt{2\pi} \sigma(M)} \exp\left(-\frac{\delta^2}{2\sigma^2(M)}\right) d\delta, \quad (14)$$

where $\sigma(M)$ is the variance of the primordial density field smoothed on the scale corresponding to PBH mass M . Because the integral is dominated by the tail of the Gaussian distribution, the PBH abundance is exponentially sensitive to the collapse threshold δ_c .

In the Relaxation-Driven Cyclic Cosmology (RDCC), the collapse threshold is enhanced by the relaxation-induced damping of super-horizon perturbations, as derived in Section 4:

$$\delta_c^{\text{RDCC}}(a) = \delta_c^{\text{std}} [1 + \beta \alpha(a)]. \quad (15)$$

Substituting this expression into the PBH abundance yields the relaxation-corrected mass fraction,

$$\beta_{\text{PBH}}^{\text{RDCC}}(M) = \exp\left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha(a))^2}{2\sigma^2(M)}\right]. \quad (16)$$

Even for modest values of $\alpha(a)$, the exponential enhancement of the effective threshold leads to a dramatic suppression of PBH formation in the visible sector.

The PBH mass is determined by the horizon mass at the time of collapse,

$$M_{\text{PBH}}(a_{\text{coll}}) \simeq \gamma M_H(a_{\text{coll}}) \propto g_*^{-1/2} T^{-2}, \quad (17)$$

where $\gamma \simeq 0.2$ is a numerical factor. Because the CPT-conjugate sector is colder by a factor $T_B/T_A \simeq 0.5$, the horizon mass in sector B is larger by a factor of four. This leads to a sectoral mass asymmetry:

$$M_{\text{PBH}}^{(B)} \simeq 4 M_{\text{PBH}}^{(A)}. \quad (18)$$

Combining the relaxation-corrected collapse threshold with the sectoral temperature asymmetry yields the PBH mass functions in the two sectors:

$$\beta_{\text{PBH}}^{(A)}(M) = \exp\left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha_A)^2}{2\sigma^2(M)}\right], \quad (19)$$

$$\beta_{\text{PBH}}^{(B)}(M) = \exp\left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha_B)^2}{2\sigma^2(M)}\right], \quad (20)$$

with $\alpha_B < \alpha_A$ due to the lower temperature of sector B. The exponential sensitivity implies

$$\beta_{\text{PBH}}^{(B)} \gg \beta_{\text{PBH}}^{(A)}, \quad (21)$$

even for small differences in α .

Three robust consequences follow:

- **Visible-sector suppression:** PBHs are strongly suppressed in sector A, consistent with observational constraints.
- **Mirror-sector enhancement:** Sector B generically contains a significantly larger PBH population.

- **Mass-spectrum shift:** PBHs in sector B are heavier due to the larger horizon mass at collapse.

In the next section, we translate these results into concrete observational predictions for gravitational waves, microlensing, and the PBH mass spectrum.

7 Observational Predictions

The relaxation-corrected collapse threshold and the sectoral PBH asymmetry derived in the previous sections lead to a number of concrete and testable predictions. These predictions arise from the interplay between the relaxation dynamics, the temperature asymmetry between the sectors, and the exponential sensitivity of PBH formation to the collapse threshold. In this section we summarize the most important observational consequences for gravitational waves, microlensing surveys, and the PBH mass spectrum.

7.1 Prediction 1: Suppressed PBH Abundance in the Visible Sector

The enhanced collapse threshold in sector A,

$$\delta_c^{(A)} = \delta_c^{\text{std}} [1 + \beta \alpha_A(a)], \quad (22)$$

leads to an exponential suppression of PBH formation in the visible sector. Even modest values of $\alpha_A(a)$ increase the threshold sufficiently to reduce the PBH abundance by many orders of magnitude. This prediction is consistent with the strong observational constraints from microlensing, the cosmic microwave background, and the stochastic gravitational-wave background, all of which limit the PBH abundance in the visible sector.

7.2 Prediction 2: Enhanced PBH Abundance in the CPT-Conjugate Sector

Because the CPT-conjugate sector is colder and experiences a smaller relaxation coefficient,

$$\alpha_B(a) < \alpha_A(a), \quad (23)$$

the collapse threshold in sector B is lower,

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (24)$$

This leads to an exponentially larger PBH abundance in sector B. While these PBHs are not directly observable, their existence is a robust prediction of the bipartite global state and may leave indirect signatures through gravitational-wave backgrounds or sectoral energy balance constraints.

7.3 Prediction 3: Sectoral Mass Asymmetry

The colder temperature of sector B implies a larger horizon mass at the time of collapse,

$$M_H^{(B)} \simeq 4 M_H^{(A)}, \quad (25)$$

leading to heavier PBHs in the CPT-conjugate sector. This mass asymmetry is a distinctive signature of RDCC and arises directly from the temperature ratio $T_B/T_A \simeq 0.5$ derived in Companion X.

7.4 Prediction 4: Gravitational-Wave Signatures

PBHs contribute to the stochastic gravitational-wave background (SGWB) through mergers and through the collapse process itself. The suppression of PBHs in the visible sector implies:

- a reduced merger rate relative to PBH-dominated dark matter scenarios,
- the absence of a PBH-induced SGWB component in the visible sector,
- potential indirect signatures from the mirror sector, depending on the degree of inter-sectoral coupling.

Future space-based interferometers may be able to detect or constrain these signatures.

7.5 Prediction 5: Microlensing Constraints

The strong suppression of PBHs in the visible sector implies that microlensing surveys such as OGLE, EROS, and Subaru-HSC should observe no significant PBH population. This prediction is consistent with current data and provides a clean observational test of the RDCC PBH sector.

7.6 Prediction 6: Modified PBH Mass Spectrum

The relaxation-corrected PBH mass function,

$$\beta_{\text{PBH}}^{\text{RDCC}}(M) = \exp \left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha(a))^2}{2\sigma^2(M)} \right], \quad (26)$$

predicts a mass spectrum in the visible sector that is:

- strongly suppressed at all masses,
- shifted toward lower masses relative to the mirror sector,
- consistent with the absence of PBH-induced anomalies in the matter power spectrum.

7.7 Prediction 7: Consistency with RDCC Temperature Asymmetry

The sectoral PBH asymmetry provides an independent test of the temperature ratio $T_B/T_A \simeq 0.5$ derived in Companion X. A precise measurement of PBH constraints, combined with CMB and large-scale-structure data, can therefore test the internal consistency of the RDCC framework.

Together, these predictions establish PBHs as a sensitive probe of relaxation-driven cosmology and provide multiple observational avenues for testing the RDCC framework.

8 Conclusion

In this Companion we have developed the primordial black hole (PBH) sector of the Relaxation-Driven Cyclic Cosmology (RDCC), closing one of the remaining major gaps in the framework. Starting from the bipartite CPT-symmetric global state and the universal relaxation mechanism introduced in Companion X, we derived the relaxation-corrected collapse threshold, the sectoral PBH asymmetry, and the resulting PBH mass functions in both sectors.

The key insight is that the relaxation dynamics fundamentally modify the early-Universe conditions relevant for PBH formation. The relaxation-induced damping of super-horizon perturbations increases the effective collapse threshold in the visible sector, leading to a strong suppression of PBH formation. At the same time, the temperature asymmetry between the

two sectors implies a lower collapse threshold and a larger horizon mass in the CPT-conjugate sector, resulting in an enhanced PBH abundance and a heavier PBH mass spectrum.

These effects arise naturally from the RDCC framework and require no additional fields, parameters, or fine-tuning. The sectoral PBH asymmetry is a direct consequence of the bipartite global state and the relaxation dynamics that govern the early Universe. The resulting predictions—including suppressed PBH formation in the visible sector, enhanced PBH formation in the mirror sector, and characteristic signatures in gravitational waves and microlensing constraints—provide multiple observational avenues for testing RDCC.

With the development of the PBH sector, RDCC now offers a coherent and internally consistent description of the thermal history, matter content, and early-Universe dynamics across both sectors. The framework continues to yield a rich set of observational signatures, making PBHs a powerful probe of relaxation-driven cosmology and a key component of the RDCC research program.

Appendix A: Derivation of the Relaxation-Corrected Collapse Criterion

In this appendix we derive the relaxation-corrected collapse threshold δ_c^{RDCC} used in the main text. The key modification relative to standard cosmology arises from the relaxation-induced damping term in the evolution equation for super-horizon perturbations.

A.1 Perturbation Equation with Relaxation

For a density contrast $\delta(a)$ in a radiation-dominated Universe, the standard evolution equation is

$$\ddot{\delta} + 2H\dot{\delta} - c_s^2 \frac{k^2}{a^2} \delta = 0, \quad (27)$$

where $c_s = 1/\sqrt{3}$ is the sound speed. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces an additional damping term proportional to the relaxation rate $\Gamma(a)$, yielding

$$\ddot{\delta} + (2H + \Gamma)\dot{\delta} - c_s^2 \frac{k^2}{a^2} \delta = 0. \quad (28)$$

During the early Universe, when $\Gamma \gg H$, the damping term dominates and suppresses the growth of super-horizon perturbations. As the Universe expands, the relaxation rate decreases according to

$$\Gamma(a) = \alpha(a) H(a), \quad (29)$$

with $\alpha(a)$ a slowly varying dimensionless coefficient.

A.2 Suppression of Super-Horizon Growth

To quantify the suppression, we consider the regime $k \ll aH$ and write the perturbation equation as

$$\ddot{\delta} + (2 + \alpha) H \dot{\delta} \simeq 0. \quad (30)$$

Assuming $H \propto a^{-2}$ during radiation domination, the solution takes the form

$$\delta(a) \propto a^{-\alpha/2}. \quad (31)$$

Thus, the relaxation-induced damping suppresses the amplitude of δ by a factor

$$S(a) = \left(\frac{a}{a_{\text{in}}} \right)^{-\alpha/2}, \quad (32)$$

where a_{in} is the scale factor at the onset of the relaxation-dominated regime.

A.3 Effective Collapse Threshold

A perturbation that would have reached the standard collapse threshold δ_c^{std} at horizon re-entry is reduced by the suppression factor $S(a)$, implying that the initial amplitude must be larger by a factor $S^{-1}(a)$. This leads to the relaxation-corrected collapse threshold

$$\delta_c^{\text{RDCC}} = \delta_c^{\text{std}} S^{-1}(a) = \delta_c^{\text{std}} \left(\frac{a}{a_{\text{in}}} \right)^{\alpha/2}. \quad (33)$$

Expanding for small α yields

$$\delta_c^{\text{RDCC}} \simeq \delta_c^{\text{std}} \left[1 + \frac{\alpha}{2} \ln \left(\frac{a}{a_{\text{in}}} \right) \right]. \quad (34)$$

Since the logarithmic factor is of order unity during the relevant epoch, we may write

$$\delta_c^{\text{RDCC}} = \delta_c^{\text{std}} [1 + \beta \alpha(a)], \quad (35)$$

with β a dimensionless coefficient of order unity determined by the perturbation profile.

A.4 Robustness of the Result

The enhancement of the collapse threshold depends only on the ratio $\Gamma/H = \alpha(a)$ and is therefore insensitive to:

- moderate variations in the temperature ratio T_B/T_A ,
- small perturbations in the relaxation coefficient $\alpha(a)$,
- higher-order corrections to the perturbation equation.

The relaxation-corrected collapse threshold is thus a robust and natural consequence of the RDCC framework and plays a central role in suppressing PBH formation in the visible sector.

Appendix B: Sectoral Temperature Effects on PBH Formation

In this appendix we analyze how the temperature asymmetry between the two sectors of the Relaxation-Driven Cyclic Cosmology (RDCC) affects primordial black hole (PBH) formation. As shown in Companion X, the temperature ratio between the sectors satisfies

$$\frac{T_B}{T_A} \simeq 0.5, \quad (36)$$

implying that the CPT-conjugate sector (sector B) is colder throughout the early Universe. This temperature difference modifies the horizon mass, the sound speed, and the collapse threshold in each sector, leading to a robust sectoral asymmetry in PBH formation.

B.1 Horizon Mass Scaling

The horizon mass during radiation domination is given by

$$M_H \propto g_*^{-1/2} T^{-2}, \quad (37)$$

where g_* is the effective number of relativistic degrees of freedom. Since sector B is colder by a factor of two, its horizon mass is larger by a factor of four:

$$M_H^{(B)} \simeq 4 M_H^{(A)}. \quad (38)$$

This implies that PBHs formed in sector B are generically heavier than those formed in sector A, even if the underlying primordial power spectrum is identical in both sectors.

B.2 Sound Speed and Collapse Dynamics

The sound speed in a relativistic plasma is

$$c_s = \frac{1}{\sqrt{3}}, \quad (39)$$

but the effective sound horizon depends on the temperature through the expansion rate. Because the colder sector expands more slowly at fixed scale factor, perturbations in sector B spend more time near the collapse threshold, enhancing the probability of PBH formation.

B.3 Temperature Dependence of the Collapse Threshold

The collapse threshold depends on the competition between gravitational attraction and pressure support. Numerical studies show that δ_c increases with temperature due to the stronger pressure in hotter plasmas. In RDCC, this implies

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (40)$$

Combined with the relaxation-induced enhancement of $\delta_c^{(A)}$ derived in Appendix A, this leads to a significant sectoral asymmetry in PBH formation.

B.4 Combined Effect on PBH Abundance

The PBH abundance in each sector is exponentially sensitive to the collapse threshold:

$$\beta_{\text{PBH}} \propto \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right). \quad (41)$$

Using the temperature dependence of δ_c and the relaxation-modified threshold from Appendix A, we obtain

$$\beta_{\text{PBH}}^{(A)} = \exp\left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha_A)^2}{2\sigma^2}\right], \quad (42)$$

$$\beta_{\text{PBH}}^{(B)} = \exp\left[-\frac{\left(\delta_c^{\text{std}}\right)^2 (1 + \beta \alpha_B)^2}{2\sigma^2}\right], \quad (43)$$

with $\alpha_B < \alpha_A$ due to the lower temperature of sector B. The exponential sensitivity implies

$$\beta_{\text{PBH}}^{(B)} \gg \beta_{\text{PBH}}^{(A)}, \quad (44)$$

even for small differences in α .

B.5 Summary

The temperature asymmetry between the two sectors leads to:

- a larger horizon mass in sector B,
- a lower collapse threshold in sector B,
- a higher PBH abundance in sector B,
- heavier PBHs in sector B,
- strong suppression of PBHs in the visible sector.

These effects are robust consequences of the RDCC framework and play a central role in the sectoral PBH asymmetry developed in the main text.

References

- [Carr and Hawking(1974)] B. J. Carr and S. W. Hawking, Mon. Not. Roy. Astron. Soc. **168**, 399 (1974).
- [Carr and Kühnel(2021)] B. Carr and F. Kühnel, Rev. Mod. Phys. **93**, 015004 (2021).
- [Sasaki *et al.*(2018)Sasaki, Suyama, Tanaka, and Yokoyama] M. Sasaki, T. Suyama, T. Tanaka, and S. Yokoyama, Class. Quant. Grav. **35**, 063001 (2018).
- [Green and Kavanagh(2021)] A. M. Green and B. J. Kavanagh, J. Phys. G **48**, 043001 (2021).
- [Collaboration(2020)] P. Collaboration, Astron. Astrophys. **641**, A6 (2020).
- [Collaboration and Collaboration(2021)] L. S. Collaboration and V. Collaboration, Phys. Rev. Lett. **126**, 241102 (2021).
- [Tisserand(2007)] P. e. a. Tisserand, Astron. Astrophys. **469**, 387 (2007).
- [Niikura(2019)] H. e. a. Niikura, Nature Astronomy **3**, 524 (2019).
- [Baumann(2020)] D. e. a. Baumann, Phys. Rev. D **102**, 023507 (2020).